GX-241 Instruction Set Documentation

Revision History		
Rev.	**Summary of Change**	**Date**
XA	Preliminary	2/2/10

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LT320103-01

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Instruction Set Command Report

Legend

Command parameter headers are light blue.
Return parameter headers are light red.
Motion command headers are light yellow.

System

Device Name	System
Version	1.2.1.0
Unit ID	-1

Command Name	Notes	Parameters
Get Instruction Set	Gets the instruction set from the connected instrument.	
Delete Instruction Set	Deletes the instruction set from the connected instrument.	
Renew Lease	Renews the lease with the connected instrument.	
Drop Lease	Drops the lease with the connected instrument.	
Get Firmware Version	Gets the Firmware version of the connected instrument.	

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Set MAC Address</td><td>Sets the new MAC address for the connected instrument.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>MAC</td><td>string</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Kill</td><td>Disconnects the specified client socket connection.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>SocketHandle</td><td>string</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Reset</td><td>Resets the Ethernet module.</td><td></td></tr>
<tr><td>Refresh Beacon</td><td>Requests beacon information from the connected instrument.</td><td></td></tr>
<tr><td>Admin</td><td>Requests the connected instrument for Admin privileges. NOTE: Application with admin privileges can execute any command on the instrument.</td><td></td></tr>
<tr><td>Get Last Response</td><td>Gets the last response sent from the instrument.</td><td></td></tr>
<tr><td>Status Rate</td><td>Sets the status rate of the instrument to the specified interval.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>Rate</td><td>number</td><td></td><td></td><td>-</td><td></td></tr><tr><td>Full Status Interval</td><td>number</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Get Client Connections</td><td>Gets all the clients information connected to the instrument.</td><td></td></tr>
<tr><td>Get Status</td><td></td><td></td></tr>
<tr><td>Flash</td><td></td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>UnitId</td><td>number</td><td></td><td></td><td>-</td><td></td></tr><tr><td>Action</td><td>string</td><td></td><td></td><td>-</td><td></td></tr><tr><td>Data</td><td>string</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Set NVM</td><td>Sets the non-volatile memory value for the specified address. See table.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>UnitId</td><td>number</td><td></td><td></td><td>-</td><td></td></tr><tr><td>Address</td><td>number</td><td></td><td></td><td>-</td><td></td></tr><tr><td>Value</td><td>string</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Get NVM</td><td>Gets the non-volatile memory value for the specified address. See table.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>UnitId</td><td>number</td><td></td><td></td><td>-</td><td></td></tr><tr><td>Address</td><td>number</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
</table>

GX-241

Device Name	GX-241
Version	1.0.4.0
Unit ID	35

Status Command	Frequency
Get Input Contacts	Sent once per command
Get Output Contacts	Sent once per command
Get 24V Contacts	Sent once per command
Get Error	Sent once per command
Get XY Position	Always on
Get Z Position	Always on
Get Injector Position	Always on

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Get Status</td><td>Gets Motor status (‘E’ for Error,
‘R’ for Running (or homing),
‘U’ for Unhomed, ‘P’ for Parked,
‘X’ Injection Valve not Installed), XYZ Position, Valve position (Load/Inject) and Error code.)</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Notes</th></tr>
<tr><td>MotorStatus</td><td>String</td><td></td></tr>
<tr><td>X Position</td><td>Number</td><td></td></tr>
<tr><td>Y Position</td><td>Number</td><td></td></tr>
<tr><td>Z Position</td><td>Number</td><td></td></tr>
<tr><td>Valve Position</td><td>String</td><td></td></tr>
<tr><td>Error</td><td>String</td><td></td></tr>
</table>
</td></tr>
<tr><td>24V Contacts</td><td>Sets the relay output contacts to either connected (1) or disconnected (0) or no change (X).</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>24V Contact 1</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
<tr><td>24V Contact 2</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
</table>
</td></tr>
<tr><td>24V Contact 1</td><td>Sets the relay output contact 1 to either connected (1) or disconnected (0) or no change (X).</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>24V Contact 1</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>24V Contact 2</td><td>Sets the relay output contact 2 to either connected (1) or disconnected (0) or no change (X).</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>24V Contact 2</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Buffered</td><td>Legacy.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>Command</td><td>String</td><td></td><td></td><td>-</td><td></td></tr></table></td></tr>
<tr><td>Clear Error</td><td>Clears the error code.</td><td></td></tr>
<tr><td>Get Valve Position</td><td></td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Valve Position</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Get 24V Contacts</td><td>Gets the relay output contacts status power on/closed(1) or power off/open(0)</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>24V Contact 1</td><td>OnOff</td><td></td></tr><tr><td>24V Contact 2</td><td>OnOff</td><td></td></tr></table></td></tr>
<tr><td>Get Output and 24V Contacts</td><td>Gets status of relay and output contacts.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Contact 1</td><td>OnOff</td><td></td></tr><tr><td>Contact 2</td><td>OnOff</td><td></td></tr><tr><td>24V Contact 1</td><td>OnOff</td><td></td></tr><tr><td>24V Contact 2</td><td>OnOff</td><td></td></tr></table></td></tr>
<tr><td>Get Error</td><td>Gets error code. See table.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Error</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Get Input Contacts</td><td>Gets status of inputs contacts.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Contact A</td><td>OnOff</td><td></td></tr><tr><td>Contact B</td><td>OnOff</td><td></td></tr><tr><td>Contact C</td><td>OnOff</td><td></td></tr><tr><td>Contact D</td><td>OnOff</td><td></td></tr></table></td></tr>
<tr><td>Get Motor Status</td><td>Gets XYZ & injector motors status.'E' for Error, 'R' for Running (or homing), 'U' for Unhomed, 'P' for Parked, 'X' Injection Valve not Installed</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>X Motor</td><td>String</td><td></td></tr><tr><td>Y Motor</td><td>String</td><td></td></tr><tr><td>Z Motor</td><td>String</td><td></td></tr><tr><td>Injector</td><td>String</td><td></td></tr></table></td></tr>
</table>

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Get Output Contacts</td><td>Gets status of output contacts.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Contact 1</td><td>OnOff</td><td></td></tr><tr><td>Contact 2</td><td>OnOff</td><td></td></tr></table></td></tr>
<tr><td>Get Encoder Position</td><td>Gets the Encoder position.Gets X, Y and Z in mm with a resolution of 0.1mm.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>X Steps</td><td>Number</td><td></td></tr><tr><td>Y Steps</td><td>Number</td><td></td></tr><tr><td>Z Steps</td><td>Number</td><td></td></tr></table></td></tr>
<tr><td>Get Injector Position</td><td>Gets the injector valve status. “R” = Running, motor in transition, “I” = Injection, Valve in Inject position, “L” = Load, Valve in Load position, “X” = Injection valve not present</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Injector Position</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Get XY Position</td><td>Gets the position of X and Y axis. X and Y Value will be in mm with a resolution of 0.01mm.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>X Position</td><td>Number</td><td></td></tr><tr><td>Y Position</td><td>Number</td><td></td></tr></table></td></tr>
<tr><td>Get Z Position</td><td>Gets the position of Z axis in mm with a resolution of 0.01mm.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Z Position</td><td>Number</td><td></td></tr></table></td></tr>
<tr><td>Home</td><td>Homes the X, Y and Z axis. Moves Z to top of travel and X, Y to the park position. NOTE: Command will clear error states.</td><td><table><tr><th>Name</th></tr><tr><td>X Motor</td></tr><tr><td>Y Motor</td></tr><tr><td>Z Motor</td></tr></table></td></tr>
<tr><td>Identify</td><td>The PROM instrument name and version.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Identity</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Immediate</td><td>Legacy.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>Command</td><td>String</td><td>%</td><td></td><td>-</td><td></td></tr></table><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Imd Return</td><td>String</td><td></td></tr></table></td></tr>
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<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Injector Position</td><td>Gets the injection valve status.
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<tr><td>Move X</td><td>Sets the new X-axis position. NOTE: This will not enable Liquid Level Detection.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr></table><table><tr><th>Name</th></tr><tr><td>X Motor</td></tr></table></td></tr>
<tr><td>Move X LLD</td><td>Sets the new X-axis position enabling Liquid Level Detection.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr></table><table><tr><th>Name</th></tr><tr><td>X Motor</td></tr></table></td></tr>
<tr><td>Move X LLD with Speed</td><td>Moves X-axis to a new position with the specified speed enabling Liquid Level Detection.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr><tr><td>X Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr></table><table><tr><th>Name</th></tr><tr><td>X Motor</td></tr></table></td></tr>
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<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Move X with Speed and Current</td><td>Moves to new X-axis position with the specified speed and drive level.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr>
<tr><td>X Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
<tr><td>X Current</td><td>Number</td><td>100</td><td>percentage</td><td>0 - 100%</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>X Motor</td></tr>
</table>
</td></tr>
<tr><td>Move XY</td><td>Set the new X-axis and Y-axis positions.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>X Motor</td></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
<tr><td>Move XY with Speed</td><td>Move to new X-axis and Y-axis position with the specified speed.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr>
<tr><td>X Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
<tr><td>Y Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>X Motor</td></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Move XY with Speed and Current</td><td>Moves to new X-axis and Y-axis positions with the specified speed and Drive level.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>X Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 162</td><td></td></tr>
<tr><td>X Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
<tr><td>X Current</td><td>Number</td><td>100</td><td>percentage</td><td>0 - 100%</td><td></td></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
<tr><td>Y Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
<tr><td>Y Current</td><td>Number</td><td>100</td><td>percentage</td><td>0 - 100%</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>X Motor</td></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Y</td><td>Moves to new Y-axis position.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Y LLD</td><td>Moves to new Y-axis position enabling Liquid Level Detection.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Y LLD with Speed</td><td>Moves to new Y-axis position with the specified speed enabling Liquid Level Detection.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
<tr><td>Y Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Y Motor</td></tr>
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</td></tr>
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<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Move Y with Speed</td><td>Moves to new Y-axis position with the specified speed.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
<tr><td>Y Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Y with Speed and Current</td><td>Moves to new Y-axis position with the specified Speed and Drive level.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Y Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 263</td><td></td></tr>
<tr><td>Y Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
<tr><td>Y Current</td><td>Number</td><td>100</td><td>percentage</td><td>0 - 100%</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Y Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Z</td><td>Moves to new Z-axis position.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Z Position</td><td>Number</td><td>0</td><td>mm</td><td>-</td><td>Based on the Z-arm used.</td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Z Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Z with Speed and Current</td><td>Moves to new Z position with the specified speed and drive level.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Z Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 125</td><td></td></tr>
<tr><td>Z Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
<tr><td>Z Current</td><td>Number</td><td>100</td><td>percentage</td><td>0 - 100%</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Z Motor</td></tr>
</table>
</td></tr>
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<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Move Z with Speed</td><td>Moves to new Z-axis position with the specified speed.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Z Position</td><td>Number</td><td>0</td><td>mm</td><td>-</td><td>Based on the Z-arm used.</td></tr>
<tr><td>Z Speed</td><td>Number</td><td>125</td><td>mm/sec</td><td>0 - 125</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Z Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Z with LLD</td><td>Moves to new Z-axis position enabling Liquid Level Detection. If liquid is encountered (on a downward movement) or air is encountered (on an upward movement) the movement stops immediately. If the position is the same as the target specified it may be assumed that liquid (or air) was not encountered</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Z Position</td><td>Number</td><td>0</td><td>mm</td><td>-</td><td>Based on the Z-arm used.</td></tr>
<tr><td>Z Speed</td><td>Number</td><td>40</td><td>mm/sec</td><td>0 - 125</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Z Motor</td></tr>
</table>
</td></tr>
<tr><td>Move Z LLD with Speed</td><td>Moves to new Z-position with the specified speed enabling Liquid Level Detection.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Z Position</td><td>Number</td><td>0</td><td>mm</td><td>0 - 125</td><td></td></tr>
<tr><td>Z Speed</td><td>Number</td><td>350</td><td>mm/sec</td><td>0 - 350</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Z Motor</td></tr>
</table>
</td></tr>
<tr><td>Output Contacts</td><td>Get the status of the relay output contacts.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Contact 1</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
<tr><td>Contact 2</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
</table>
</td></tr>
<tr><td>Output Contact 1</td><td>Gets the status of relay output contact 1.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Contact</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
</table>
</td></tr>
<tr><td>Output Contact 2</td><td>Get the status of relay output contact 2.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Contact</td><td>OnOff</td><td>On</td><td></td><td>-</td><td></td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Pulse Contact</td><td>Sets the specified relay output contact for pulse time specified. For example values 1 and 0.5 sets the relay contact 1 to be closed for 0.5 seconds.</td><td><table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Contact</td><td>Number</td><td>1</td><td></td><td>-</td><td>1 implies contact relay 1 which is second from bottom, 2 implies contact relay 2 which is bottom one.</td></tr>
<tr><td>Time</td><td>Number</td><td>0.1</td><td>secs</td><td>-</td><td></td></tr>
</table></td></tr>
<tr><td>Wait For X</td><td>Wait for X-Motor to come to parked position.</td><td><table>
<tr><th>Name</th></tr>
<tr><td>X Motor</td></tr>
</table></td></tr>
<tr><td>Wait For XY</td><td>Wait for X and Y Motors to come to parked position.</td><td><table>
<tr><th>Name</th></tr>
<tr><td>X Motor</td></tr>
<tr><td>Y Motor</td></tr>
</table></td></tr>
<tr><td>Wait For Y</td><td>Wait for Y-Motor to come to parked position.</td><td><table>
<tr><th>Name</th></tr>
<tr><td>Y Motor</td></tr>
</table></td></tr>
<tr><td>Wait For Z</td><td>Wait for Z-Motor to come to parked position.</td><td><table>
<tr><th>Name</th></tr>
<tr><td>Z Motor</td></tr>
</table></td></tr>
</table>

GX Syringe Pump

Device Name	**GX Syringe Pump**
Version	1.0.1.0
Unit ID	0

Status Command	**Frequency**
Get Error	Sent once per command
Get Syringe Info	Always on

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Aspirate</td><td>Flips valve to position specified and then aspirates specified volume with the specified flow rate.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Valve Position</td><td>String</td><td>N</td><td></td><td>-</td><td>'N' for needle 'R' for reservoir position</td></tr>
<tr><td>Volume</td><td>Number</td><td>0</td><td>µL</td><td>-</td><td>Volume in µL to aspirate</td></tr>
<tr><td>Flow Rate</td><td>Number</td><td>10</td><td>mL/min</td><td>-</td><td>Flow rate in mL/min at which to aspirate</td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Syringe</td></tr>
</table>
</td></tr>
<tr><td>Buffered</td><td>Legacy.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Command</td><td>String</td><td></td><td></td><td>-</td><td></td></tr>
</table>
</td></tr>
<tr><td>Clear Error</td><td>Clears the error code.</td><td></td></tr>
<tr><td>Dispense</td><td>Flips valve to position specified and then dispenses specified volume with the specified flow rate.</td><td>
<table>
<tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr>
<tr><td>Valve Position</td><td>String</td><td>N</td><td></td><td>-</td><td>'N' for Needle or 'R' for Reservoir position</td></tr>
<tr><td>Volume</td><td>Number</td><td>0</td><td>µL</td><td>-</td><td></td></tr>
<tr><td>Flow Rate</td><td>Number</td><td>10</td><td>mL/min</td><td>-</td><td></td></tr>
</table>
<table>
<tr><th>Name</th></tr>
<tr><td>Syringe</td></tr>
</table>
</td></tr>
</table>

<table>
<tr><th>Command Name</th><th>Notes</th><th>Parameters</th></tr>
<tr><td>Get Error</td><td>Gets error code. See table.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Error</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Get Motor Status</td><td>Get the status of Valve motor and Syringe motor status. 'U' for unpowered, 'P' for parked, 'R' for running, 'E' for error.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Syringe</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Get Syringe Info</td><td>Gets the Valve position of the syringe and the Volume of the syringe contents. Default unit for volume is µL.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Valve Position</td><td>String</td><td>Valve position can be 'R' for Reservoir, ‘N’ for Needle, ‘O’ for valve motor overload, ‘X’ for running</td></tr><tr><td>Volume</td><td>Number</td><td></td></tr></table></td></tr>
<tr><td>Halt Syringe</td><td>Halt the syringe and generate error.</td><td><table><tr><th>Name</th></tr><tr><td>Syringe</td></tr></table></td></tr>
<tr><td>Home</td><td></td><td><table><tr><th>Name</th></tr><tr><td>Syringe</td></tr></table></td></tr>
<tr><td>Identify</td><td>The PROM instrument name and version.</td><td><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Identity</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Immediate</td><td>Legacy.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>Command</td><td>String</td><td>%</td><td></td><td>-</td><td></td></tr></table><table><tr><th>Name</th><th>Type</th><th>Notes</th></tr><tr><td>Response</td><td>String</td><td></td></tr></table></td></tr>
<tr><td>Valve Position</td><td>Moves valve to specified position or stops the syringe - 'L' for Load position, 'I' for Inject position, 'X' for stop syringe.</td><td><table><tr><th>Name</th><th>Type</th><th>Default</th><th>Units</th><th>Range</th><th>Notes</th></tr><tr><td>Valve Position</td><td>String</td><td>N</td><td></td><td>-</td><td></td></tr></table><table><tr><th>Name</th></tr><tr><td>Syringe</td></tr></table></td></tr>
<tr><td>Wait For Syringe</td><td>Wait for Syringe pump to finish.</td><td><table><tr><th>Name</th></tr><tr><td>Syringe</td></tr></table></td></tr>
</table>

Non-Volatile Memory Map

GX-241

Address	***Meaning***	***Units***	***Decimal Places***	***Default***
0	Reserved			
1	Injector 0 = No Injection Valve 1 = Injection Valve	NA	NA	0
2	Safety Stop Switch Functionality Safety Input configuration: 0000 (1)000 = Option 0000 0(1)00 = Remote/Start 0000 00(1)0 = Pause/Resume 0000 000(1) = Sync (1)000 0000 = Relay Contact 1(Top) 0(1)00 0000 = Relay Contact 2 00(1)0 0000 = 24Vdc Relay 1 000(1) 0000 = 24Vdc Relay 2 (Bottom) Relay State “1” = closed; “0” = open	NA	0	00000000b
3	Park position X coordinate	mm	3	0
4	Park position Y coordinate	mm	3	0
5	Clamp height of the Z drive in mm. The minimum value is 110 mm.	mm	3	125.000
6	Reserved	-	0	0
7	Liquid sensitivity	%	2	1
10 - 27	Reserved	-	-	-
28	X offset	mm	3	0
29	Y offset	mm	3	0
30	Z offset	mm	3	0
32	Reserved			
33..36	Serial Number	ASCII		16 Chars
37..46	Machine Name	ASCII		40 Chars

GX-Syringe Pump

Address	***Meaning***	***Units***	***Decimal Places***	***Default***
0..3	Reserved	none	0	0
4	syringe size	Contains syringe size in µL as an integer. Permitted values: 100 = 100 µL syringe 250 = 250 µL syringe 500 = 500 µL syringe 1000 = 1 mL syringe 5000 = 5 mL syringe 10000 = 10 mL syringe	0	10000
5	Reserved	none	0	0
6	Backlash	uStep	0	60
7..31	Reserved	none	0	0

Error Messages

GX-241

Error Number	***Error Text***
0	No Error
10	Bad Command Entered
11	Invalid NV-RAM Address
12	Safety Switch Activated
13	Bad Parameter Entered
14	FIFO Full
15	FIFO Add
16	Character Limit
17	Park Location
20..23	[X,Y,Z,I] Axis unHomed
24..27	[X,Y,Z,I] Axis Moving
28.30	[X,Y,Z] Axis Stall
32..34	[X,Y,Z] Encoder
36..38	[X,Y,Z] Speed Range
40..42	[X,Y,Z] Target Range
99	Accessory Error

GX Syringe Pump

Error Number	Error Text
0	No error
10	Unknown buffered command
11	Invalid NV-RAM address
12	Emergency stop activated
20	Pump command while unhomed
22	Pump command while busy
24	Invalid syringe position
26	Invalid syringe volume
28	Invalid flow rate
30	Invalid syringe size
32	Invalid valve position
34	Missing valve encoder